batmonitor + opportunity loads (PROPOSAL, not built) - one cycle every 5 s
blue = decision (bits + state, make check) · green = amount · orange, thick border = new · dashed = data only

inputs (bm_parse)
PCC, Sofar Bat1, PV, heat pump W, BMS SoC / U / CCL / DCL / cells, AC-in / AC-out per phase

forecast inputs
fc_slots 24 x 3 h OWM (3 days), fc_corr, fc_ts
NEW: measured daily base load (MQTT from wagodb)

NEW table BM_LOADS (like BM_BANKS)
name · priority · nominal W · AC phase or DC bank · actuator
soc_on / soc_off · allow_grid · min_run_s · bridge_s

NEW load inputs per load
relay read-back, measured power (W), age of the feedback
DC load: AC-in x eta - BMS /Dc/0/Power of its bank

SoC

D7 house on AC-out1 (Z1 / Z2)

D5 DO4 pulse (FoxESS shedding)

NEW OPL threshold (amount layer)
E_PV(next N days) - base load x h - (TARGET_SOC - SoC) x capacity_wh
-> soc_release = clamp(MIN_RESERVE ...)
reason FC_GOOD / FC_POOR / FC_STALE -> OPL_SOC_FIXED

D1 source DAY / B1 / STK / WPG, winter heat pump cap

S0 caps per phase: chg_cap, dis_cap (charger, CCL / DCL), top hold

D4 summer peak window (export cap)

D2 mode per stack, first match
BMS_MISSING > FORCE > STALE > LADESPERRE > CHARGE / FULL > HOLD > PROT > CHARGING > DIS

chg_cap / dis_cap

soc_ge_release

no_comm

read-back next cycle

relay, W

threshold + reason

D-lead SoC balancing

mode matrix BM_MODE_RULE

stack modes, surplus

DIS

CHARGE / FORCE

NEW load layer chain_loads (between chain_mode and chain_charge)

budget = surplus - stack priority share
per AC phase: 5000 VA - AC-out load (INV_LIMIT)
allocation by priority, water filling as alloc_levels

timers per load (state: on_since, deficit_since)
min_run_over = now - on_since ≥ min_run_s
bridge_over = deficit for ≥ bridge_s, reset when surplus returns

no_power, prio_other, inv_limit

min_run_over, bridge_over

D8 per load, first match -> req + reason
1 NO_COMM (grey) · 2 SOC_FLOOR · 3 OVERLOAD / INV_LIMIT
4 MIN_RUN (green) · 5 BRIDGE (green, battery / grid if allow_grid)
6 SOC_BELOW · 7 CHARGE_FIRST · 8 PRIO_OTHER · 9 NO_POWER (red)
10 PROBE (force_full / top hold) · 11 ON (green)
power off only if min_run_over AND bridge_over; 1-3 switch at once
make check: all bit x state combinations, fixed point after one step

BRIDGE

granted W

req

time left

discharge branch (changed)
BM_SRC_RULE, D3 PROP / IDLE, alloc_discharge
BRIDGE: discharge only above OPL_BAT_SUPPORT_MIN_SOC,
below it without allow_grid -> the load is shed

charge path (changed)
want = own + KP x (surplus - target) **- granted loads**
alloc_charge up to chg_cap

EXT_LOCK (observation, not a decision)
req 1 + relay 1 + power ~0 for X s
= locked by the device (thermostat)
no power meter -> UNKNOWN (grey)

req

setpoints per phase -> Multis
rule text per phase -> state.json -> wagodb.pv_victron

NEW actuators (batmonitor.c)
GX relay /Relay/n/State · MQTT relay (pv_relay pattern) · Orion remote via relay
fail-safe: restart / STALE / BMS missing -> all loads off

NEW state.json "loads" + MQTT batmonitor/loads (retained)
per load: request · actual · reason · colour · min run / bridge time left
threshold: soc_release + its reason

NEW OPL status page
settings-web (nginx in the GX) or gui-v2 page (Remote Console)
green = released · red = blocked · grey = no communication / unknown
EXT_LOCK shown apart from the energy decision